

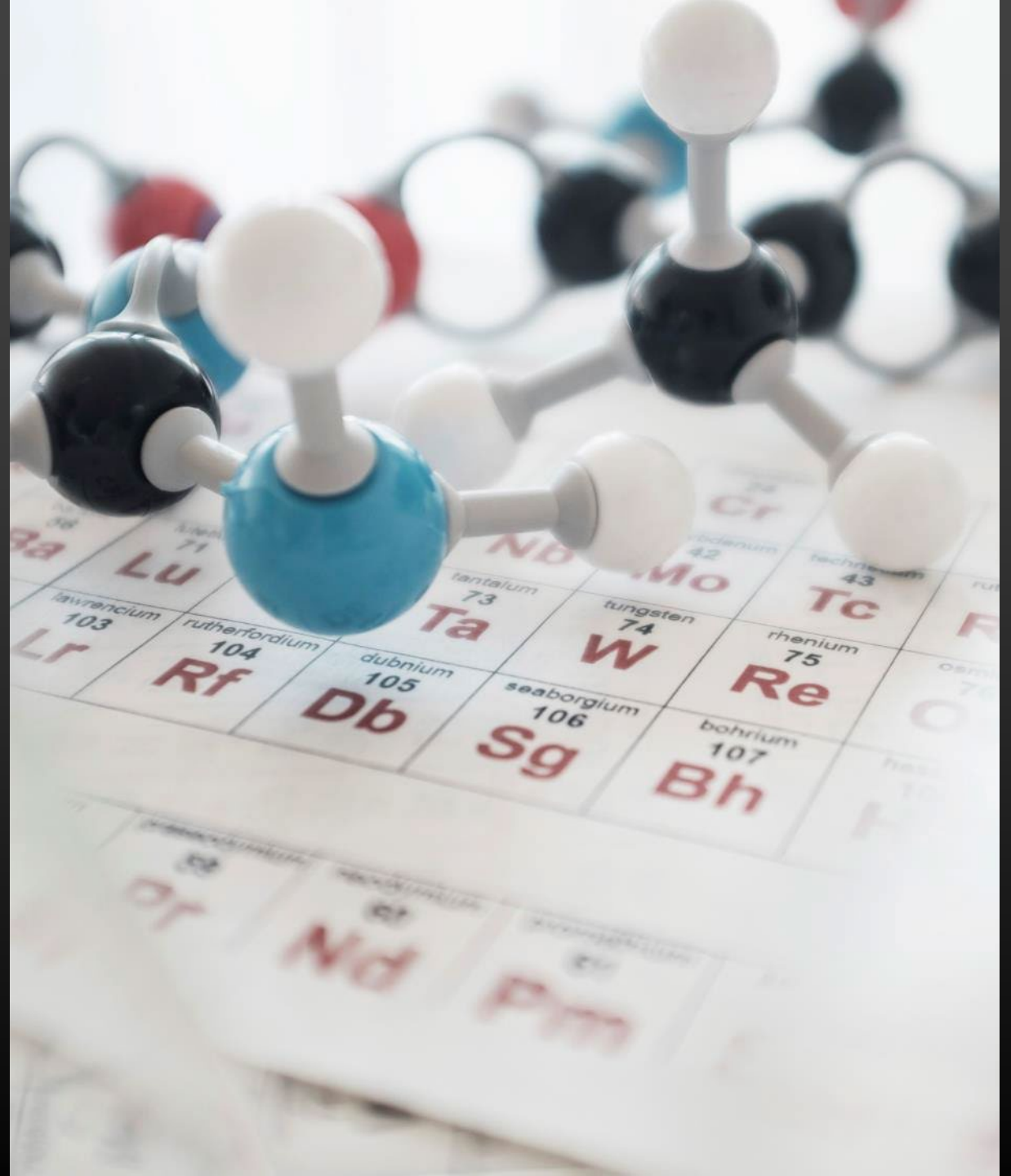
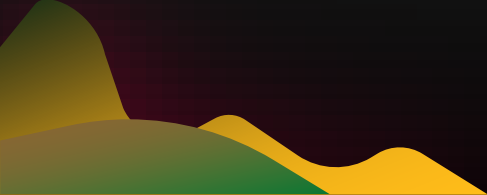


Glutamate

Synthesis, release &
inactivation

Glutamate receptors

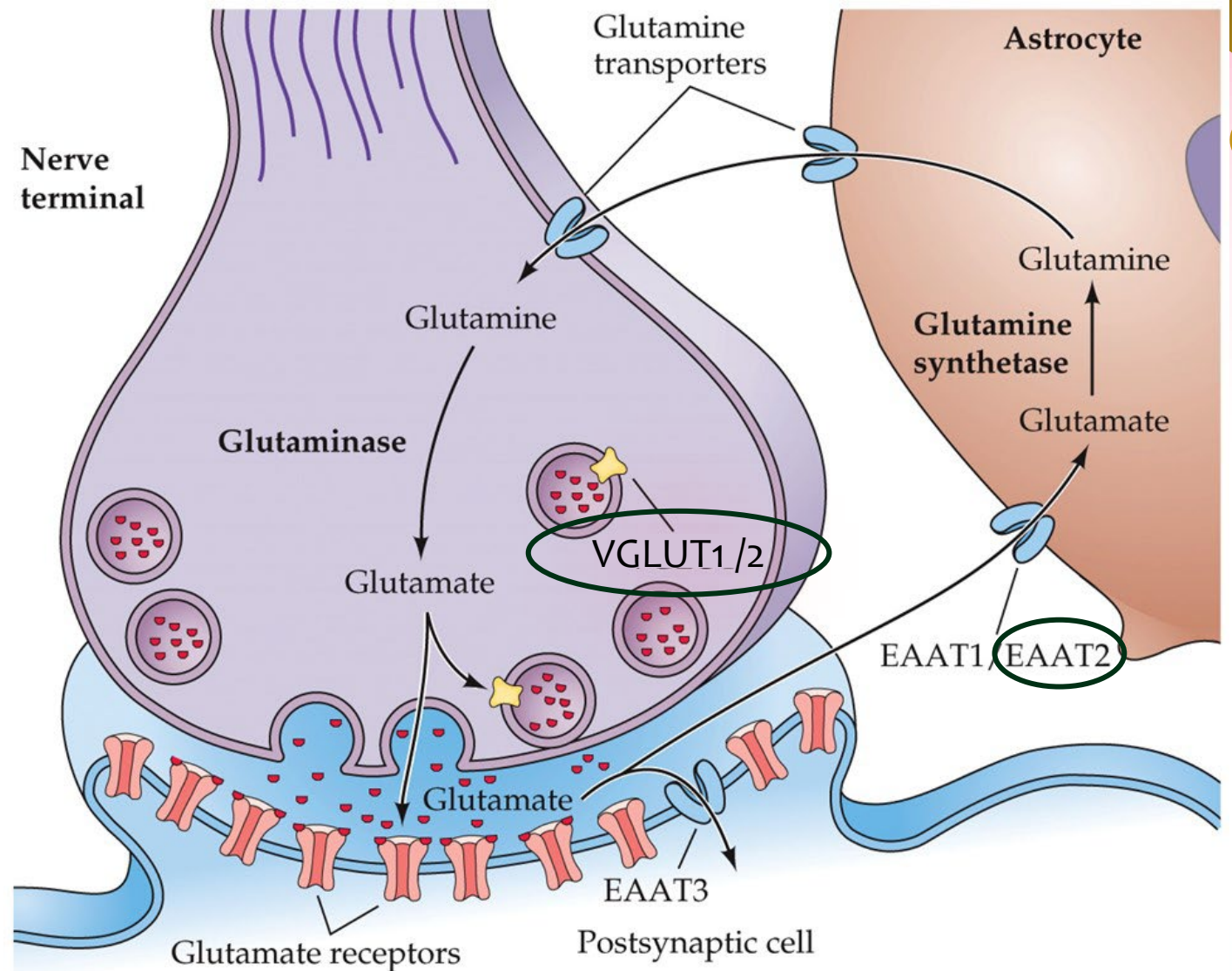
Neuronal plasticity / LTP





GLUTAMATE

Synthesis, storage & release, and inactivation



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Glutamate is an agonist of both ionotropic and metabotropic receptors

iGlu receptors

- NMDA
- non-NMDA
 - AMPA
 - kainic acid (KA)

mGlu receptors

- Group I (mGlu1, 5)
- Group II (mGlu 2, 3)
- Group III (mGlu4, 6 & 7 & 8)



iGlu receptors require 4 subunits (tetrameric)

AMPA receptors

- a “dimer of dimers.”
- GluA1 dimer together with GluA2 dimer

NMDA receptors

- GluN1 / GluN2 dimer of dimers
- glu binding site on GluN2
- gly binding site on GluN1

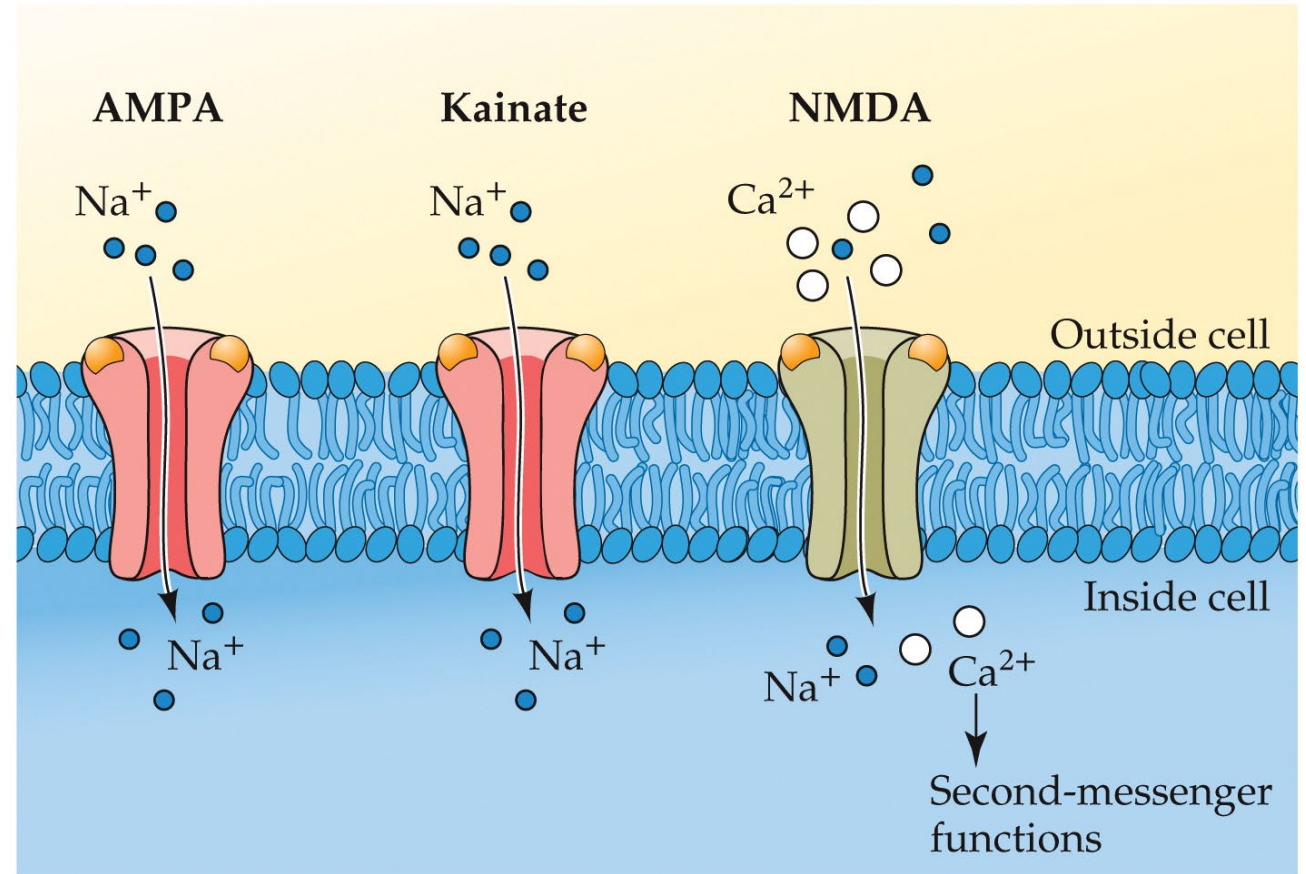
AMPA	NMDA
GluA1	GluN1
GluA2	GluN2A
GluA3	GluN2B
GluA4	GluN2C
	GluN2D
	GluN3A
	GluN3B



All 3 iGlu receptors conduct Na^+

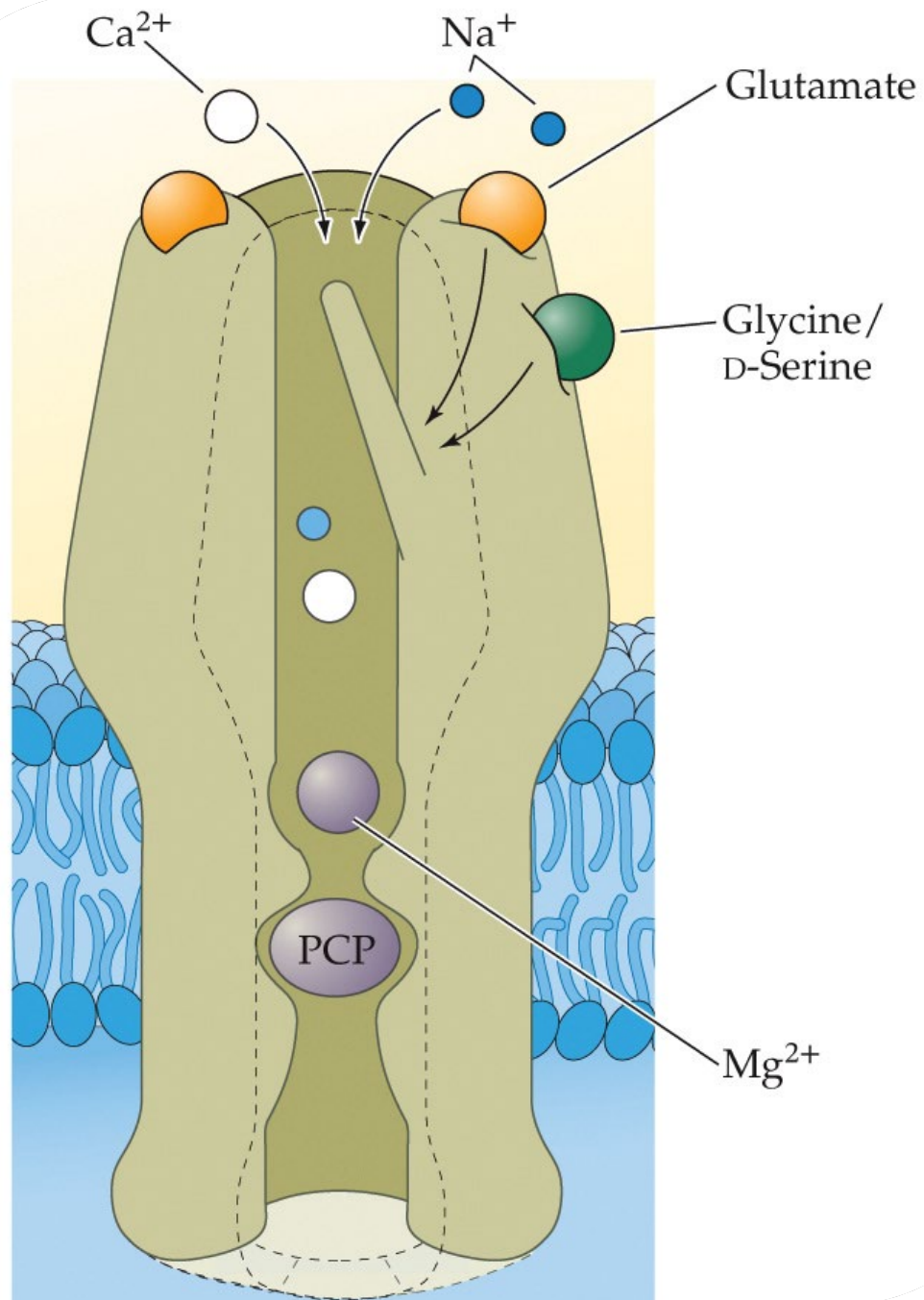
Some receptors are also permeable to Ca^{2+}

- NMDA receptor
- AMPA receptors lacking GluA2



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NMDA receptors have unusual properties

- Ligand-gated ion channels
 - Require co-agonist gly and glu
- Voltage-gated
 - Removal of Mg²⁺ block



Voltage-dependence of NR is due to Mg^{2+} block in the pore of the channel

At rest or Hyperpolarized V_m

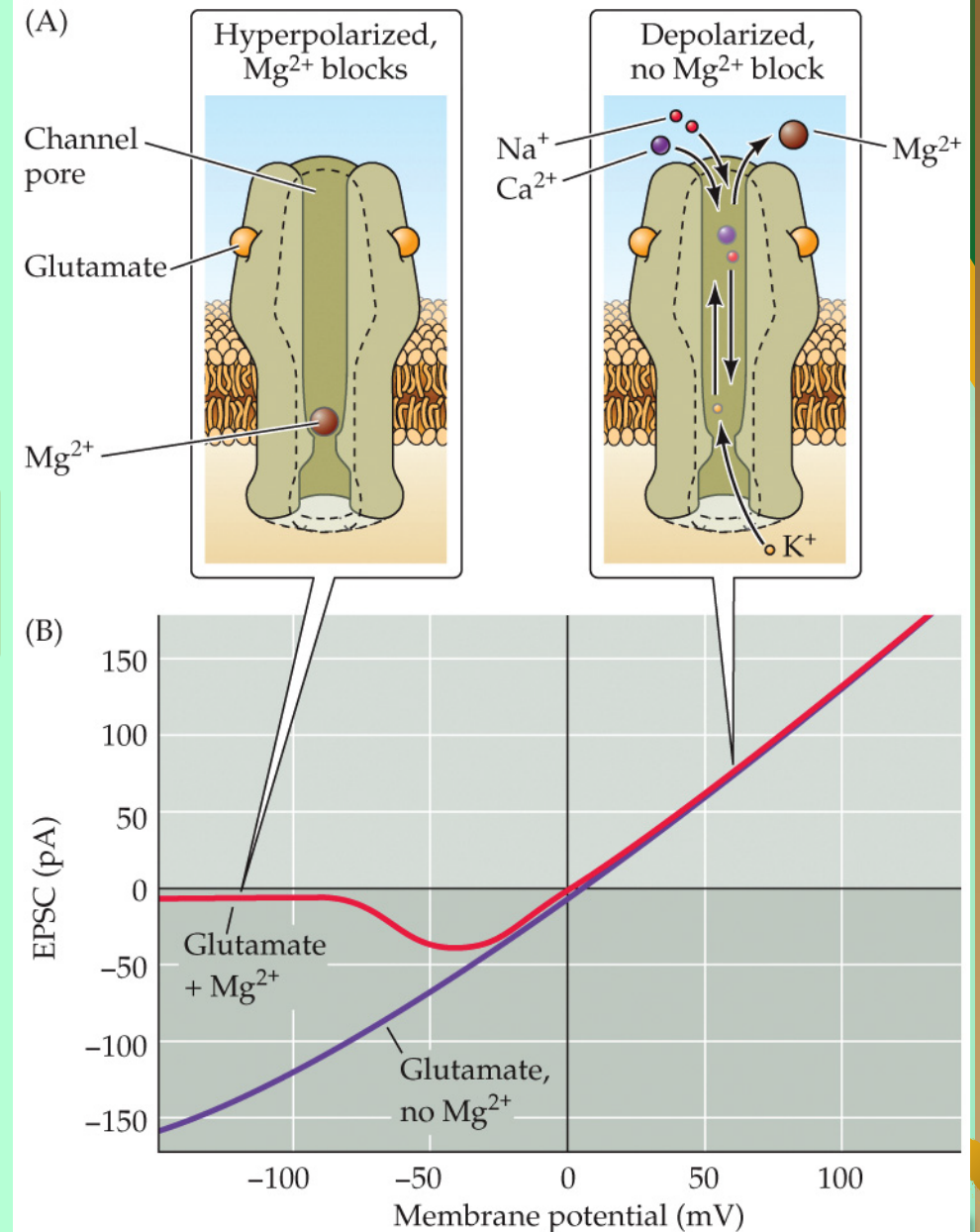
Mg^{2+} blocks the channel

Non-conductive

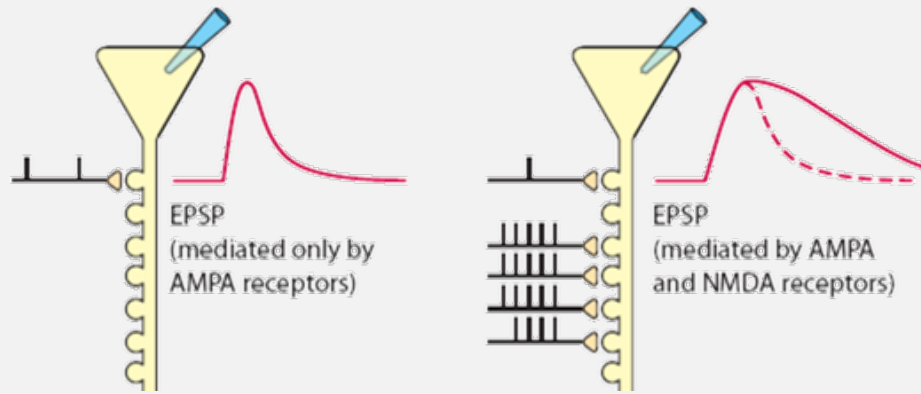
Depolarized V_m

Mg^{2+} is expelled

Highly permeable to Ca^{2+}



The functional roles of Glu are important in neurotransmission, plasticity & disease

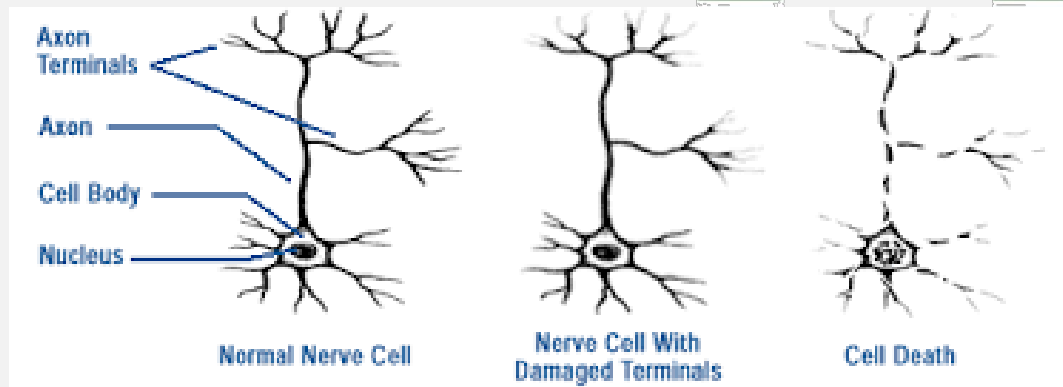
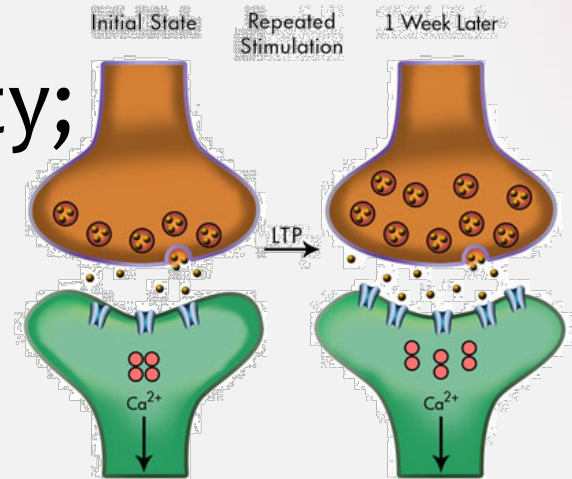


Fast excitatory neurotransmission

Excitotoxicity

Neurological disorders; neurodegeneration; stroke

Neuroplasticity; LTP



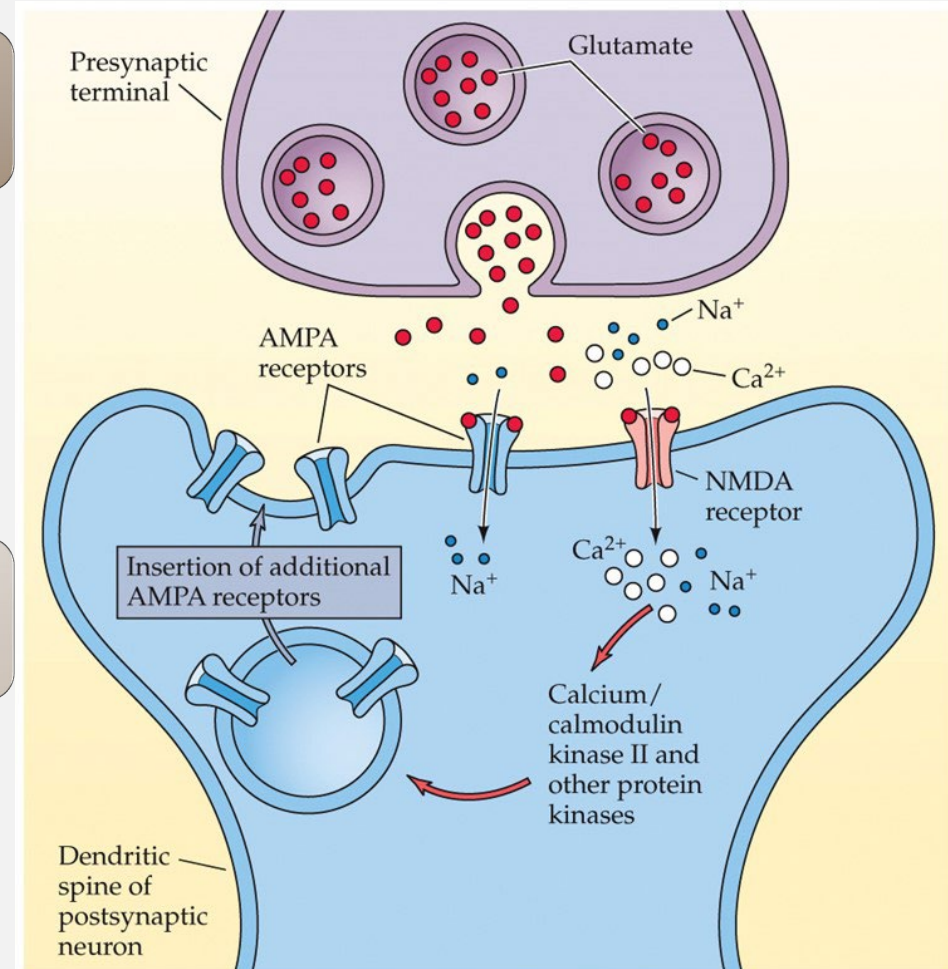
NMDA receptors are critical in some forms of long-term potentiation (LTP)

Induction phase

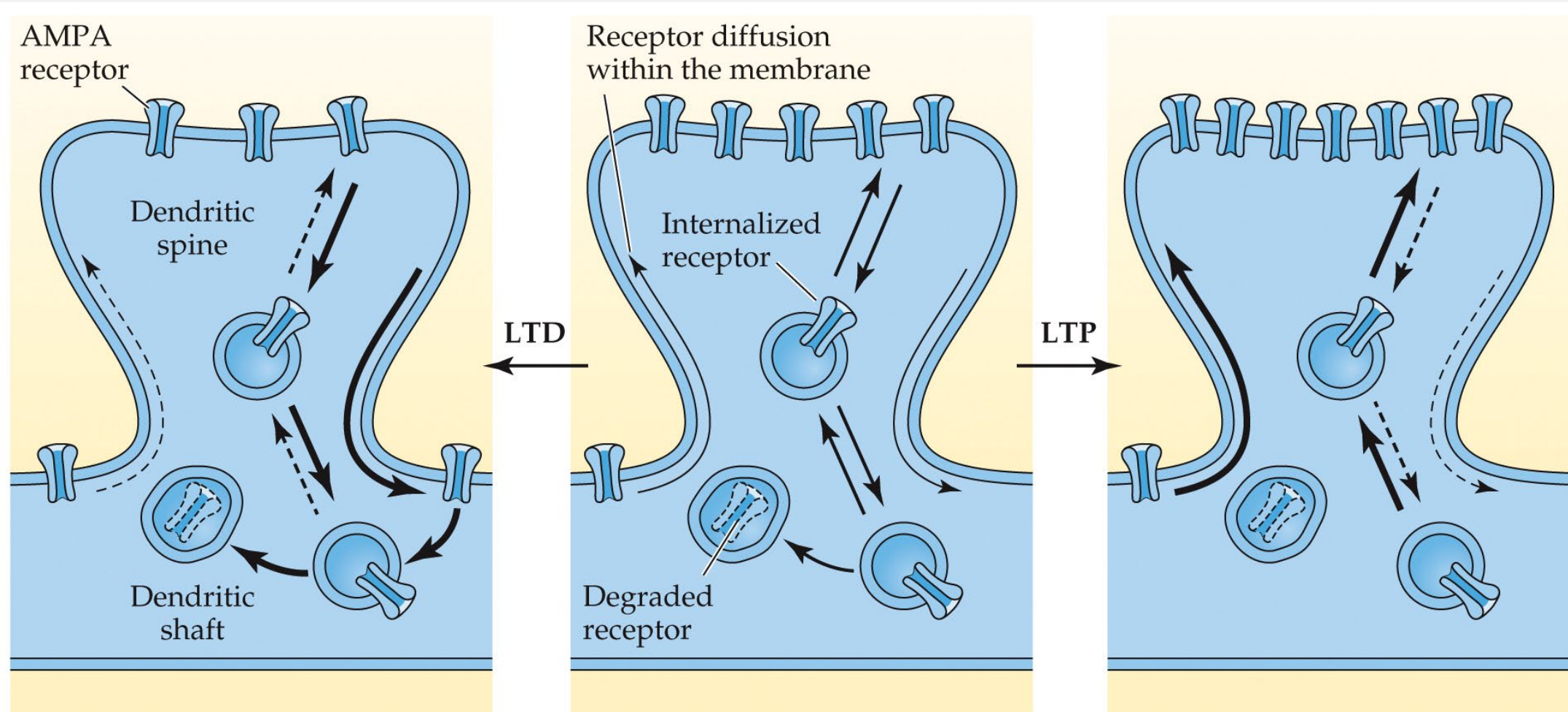
- Depolarization
- NMDA receptor activation
- Ca^{2+} influx

Expression phase

- Many kinases phosphorylate several targets involved in AMPA receptor trafficking



AMPA receptors trafficking in spines can be altered by synaptic plasticity

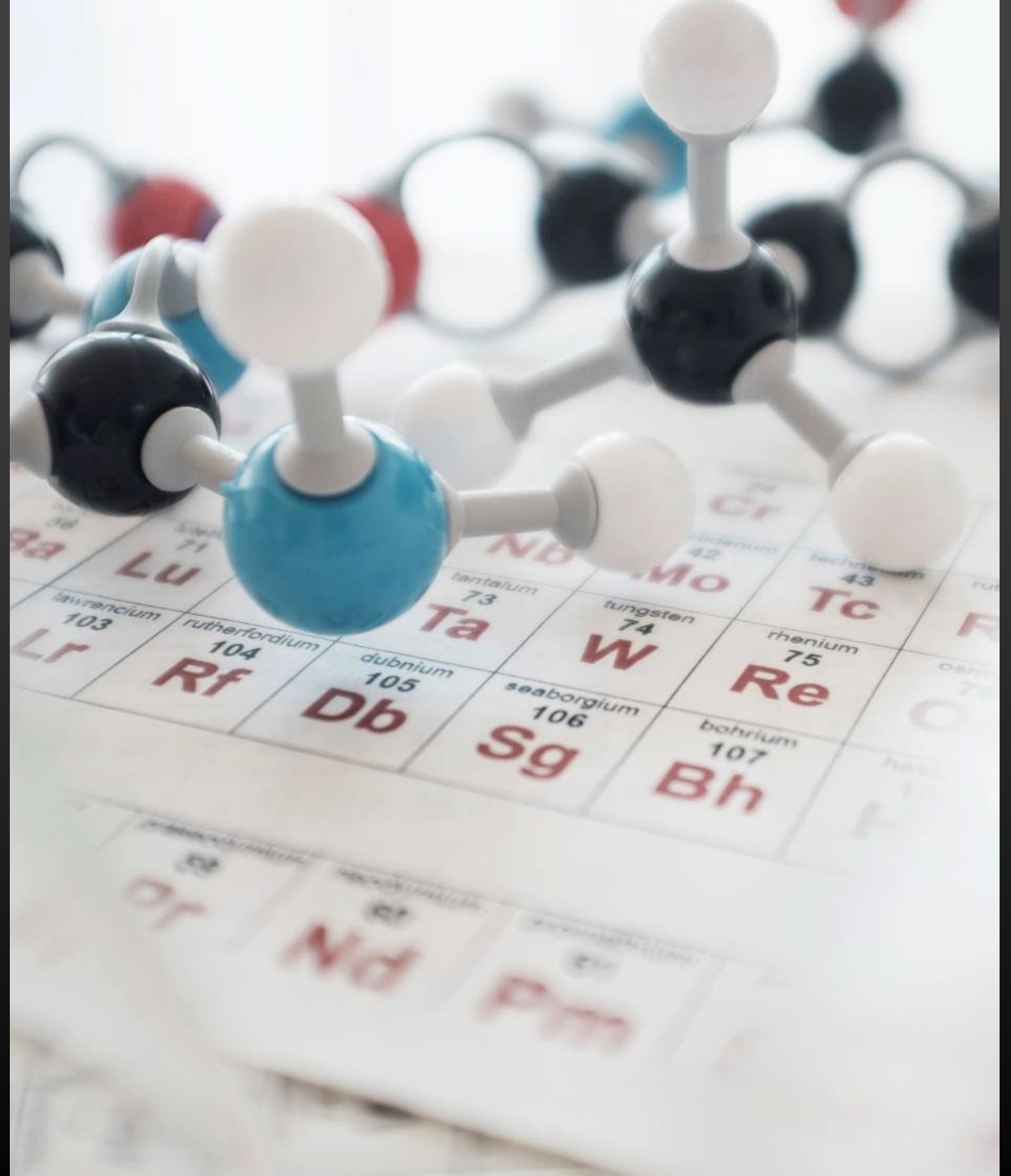
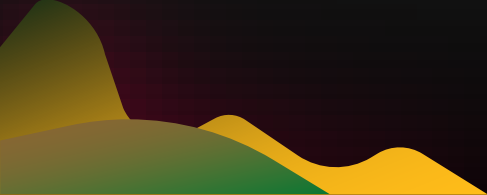




GABA

Synthesis, release &
inactivation

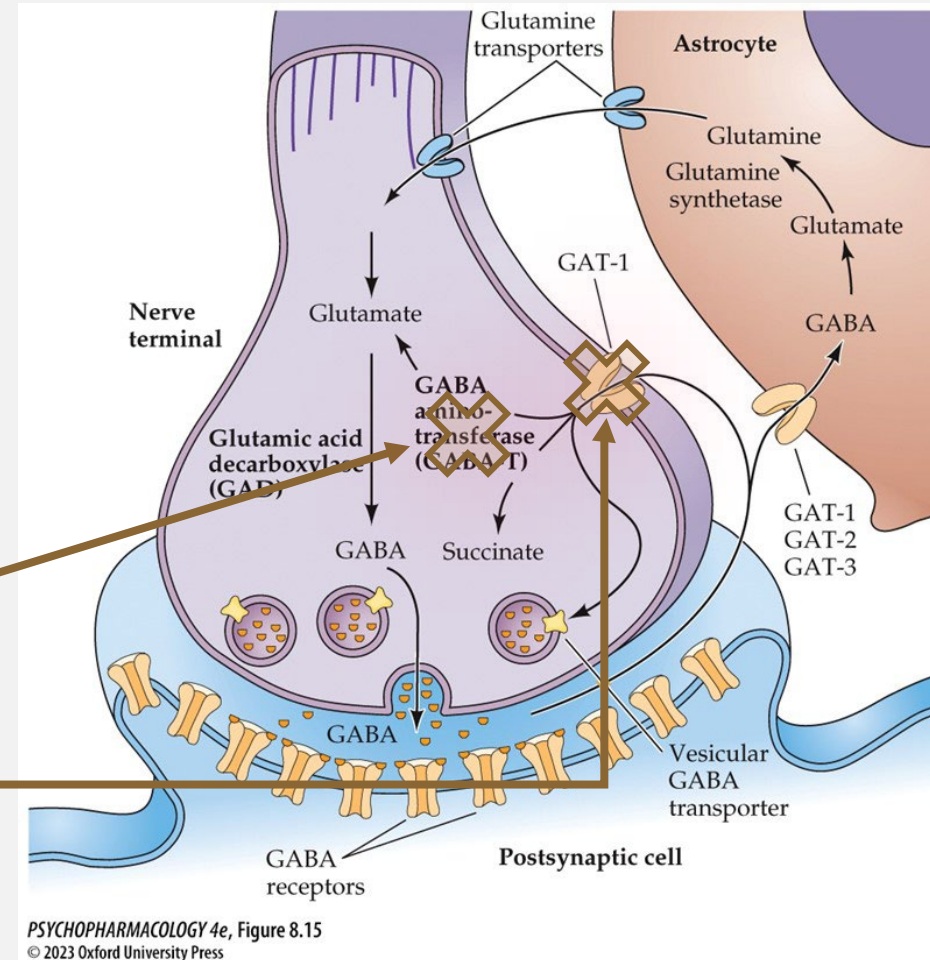
GABA_A receptors



GABA is the most abundant inhibitory neurotransmitter in the brain

CNS depressants

- Anxiolytics
- Anesthetics
- Sedatives
- Anticonvulsants
 - Vigabatrin (Sabril)
 - Tiagabine (Gabitril)

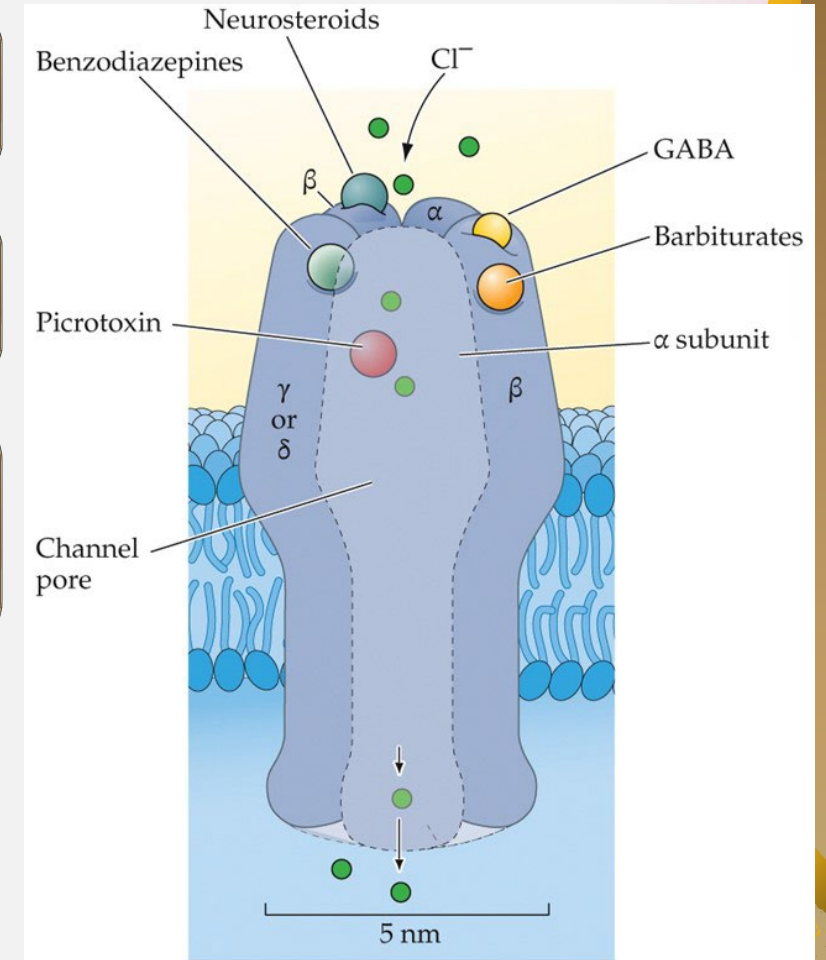


GABA_A ionotropic receptors are permeable to Cl⁻, causing hyperpolarization

5 subunits

GABA binding site

Binding sites for several **positive** and **negative** allosteric modulators

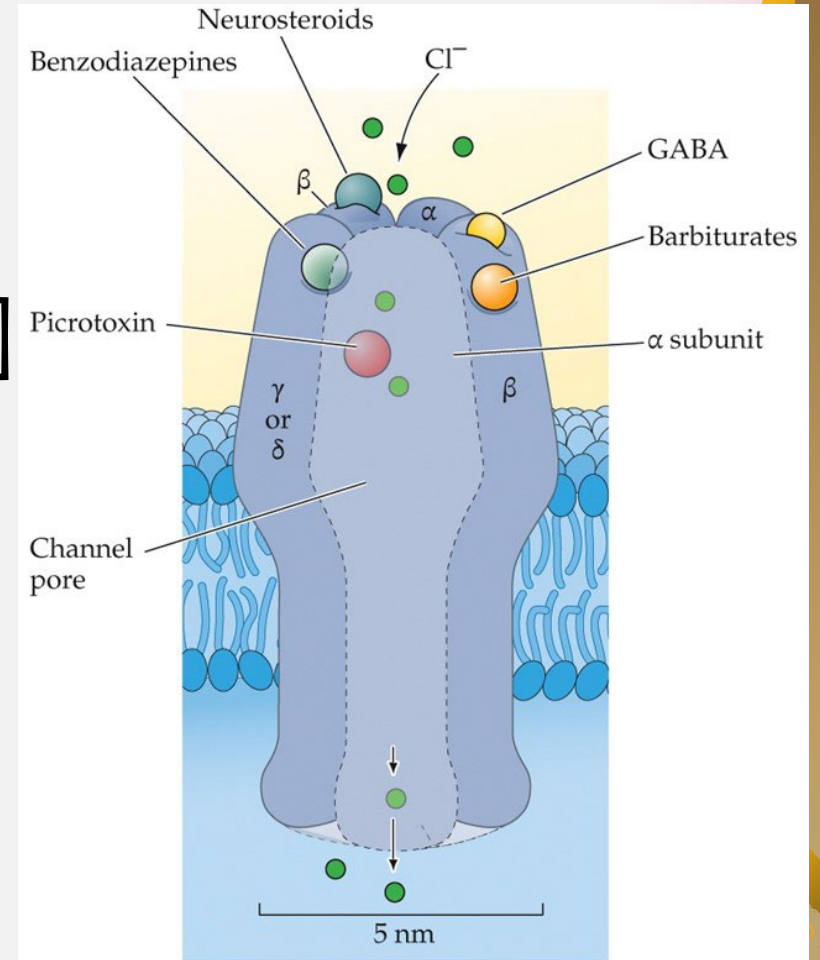


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GABA_A ionotropic receptors are permeable to Cl⁻, causing hyperpolarization

5 subunits

- Two α ; two β ; one γ [$(\alpha 1)_2, (\beta 2)_2, (\gamma 2)$]
- Extrasynaptic R's [$(\alpha 4)_2$ or $(\alpha 6)_2, (\beta_x)_2, (\delta)$]



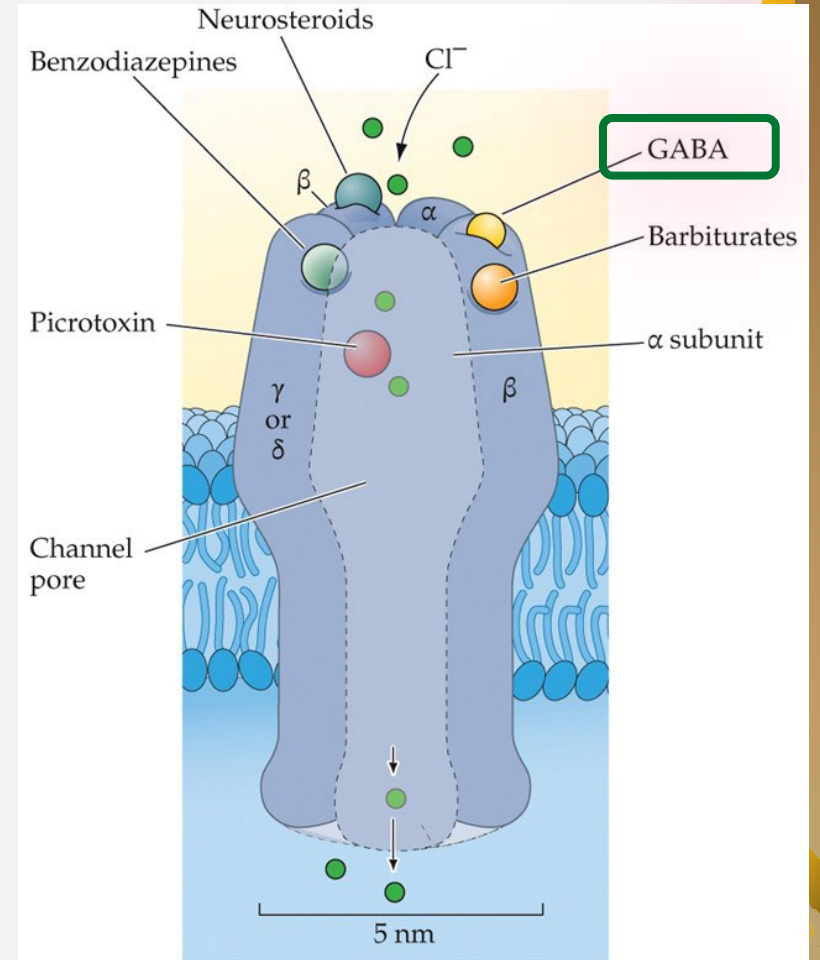
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GABA_A ionotropic receptors are permeable to Cl⁻, causing hyperpolarization

5 subunits

GABA binding site

- on the β subunit
- muscimol (agonist)
- bicuculline (competitive antagonist)



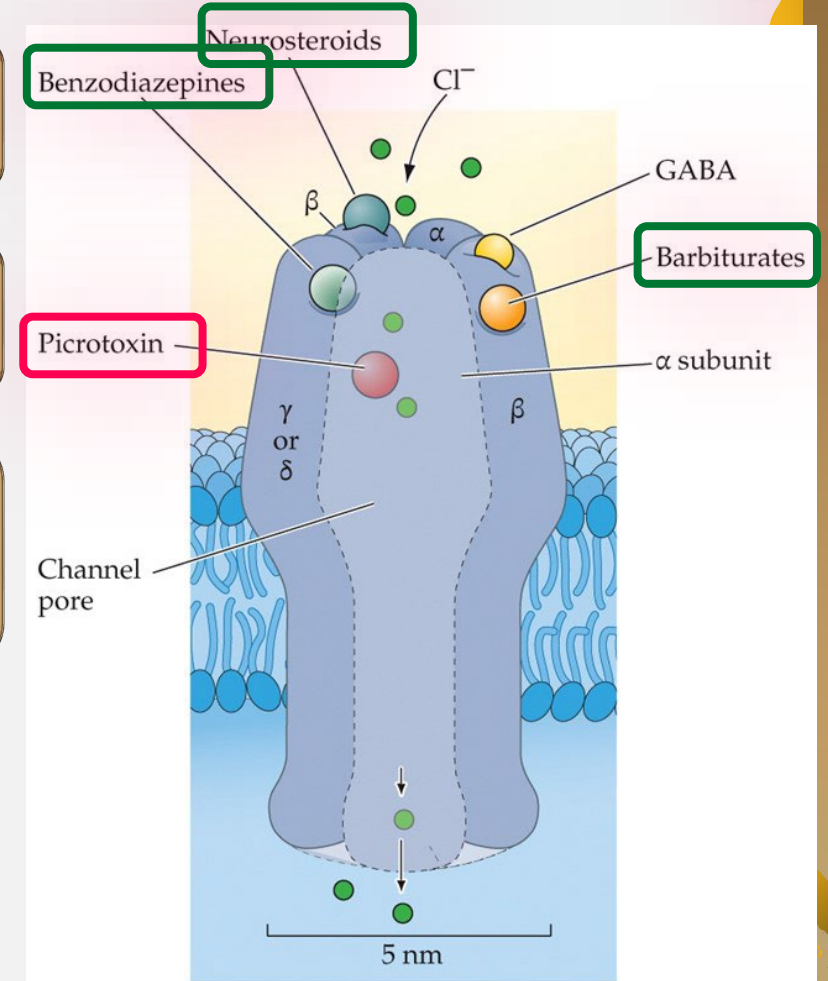
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GABA_A ionotropic receptors are permeable to Cl⁻, causing hyperpolarization

5 subunits

GABA binding site

Binding sites for several **positive** and **negative** allosteric modulators



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Many CNS depressants are (+) allosteric modulators of GABA_A receptors

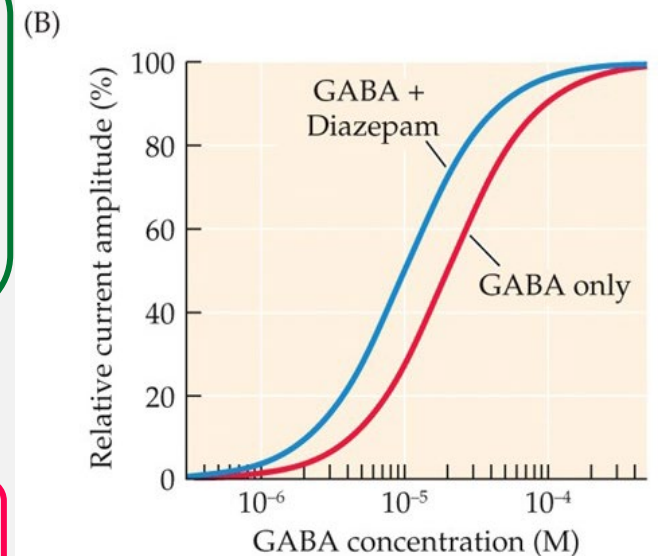
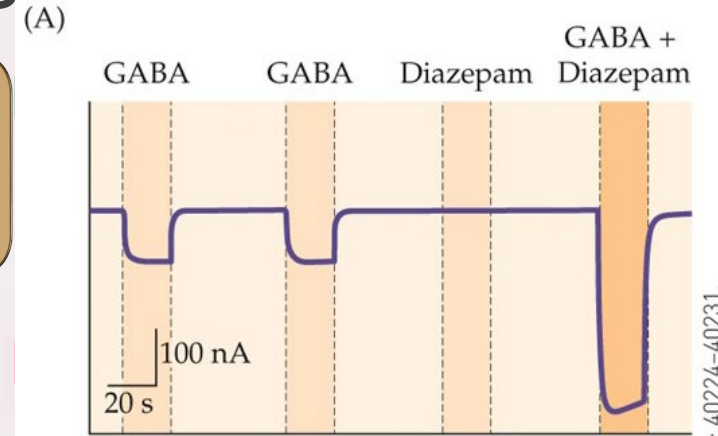
Binding sites for several **positive** and **negative** allosteric modulators

(+) PAM

- Benzodiazepines
- Barbiturates
- Neurosteroids, EtOH, & Anesthetics

(-) NAM

- Picrotoxin / pentylenetetrazol (PTZ)



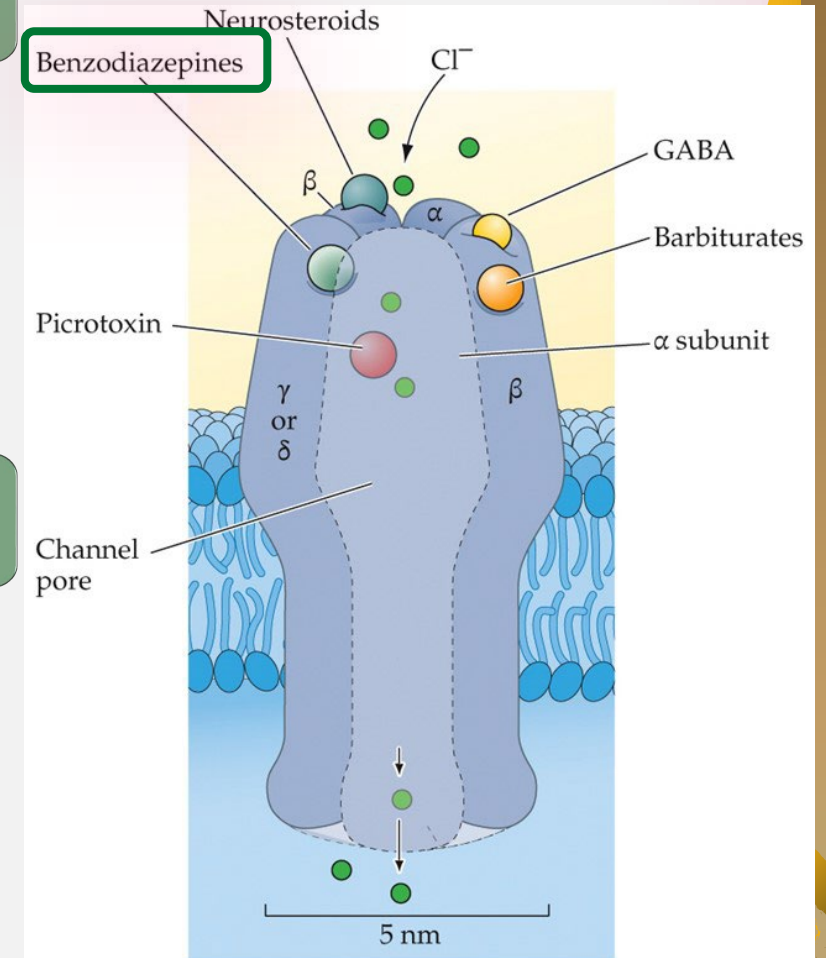
BDZ binding requires γ_2 subunit (α - γ interface)

Increases the potency of GABA

- $\uparrow$ affinity of GABA for β -subunit
- $\uparrow$ probability channel will open
- $\uparrow$ Cl^- influx ($\uparrow$ hyperpolarization)

Behavioral effects due to α -subunit

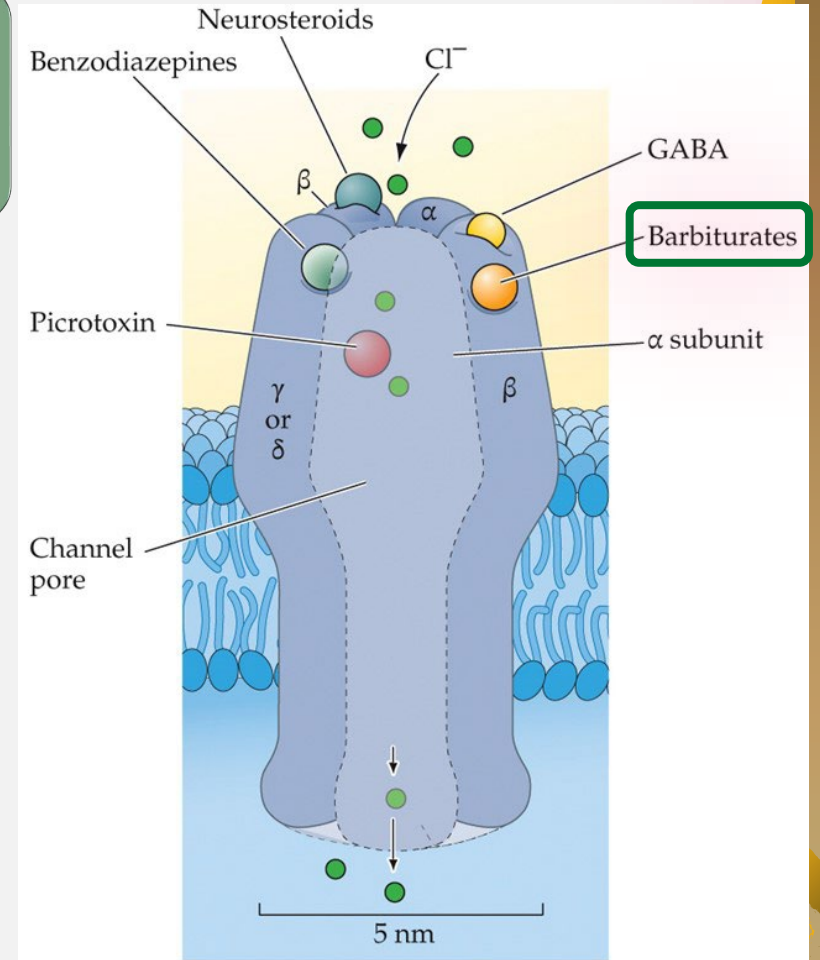
- α_1 – sedation (zolpidem/Ambien)
- α_2/α_3 – anxiolytic (alprazolam/Xanax)
- α_5 – amnestic



The barbiturate binding site

is on the β -subunit distinct from the GABA / agonist site

- $\uparrow$ Cl^- flux
- $\uparrow$ channel open probability
- $\uparrow$ duration of channel opening
- even in the absence of agonist



Other PAM's of GABA_A receptors include

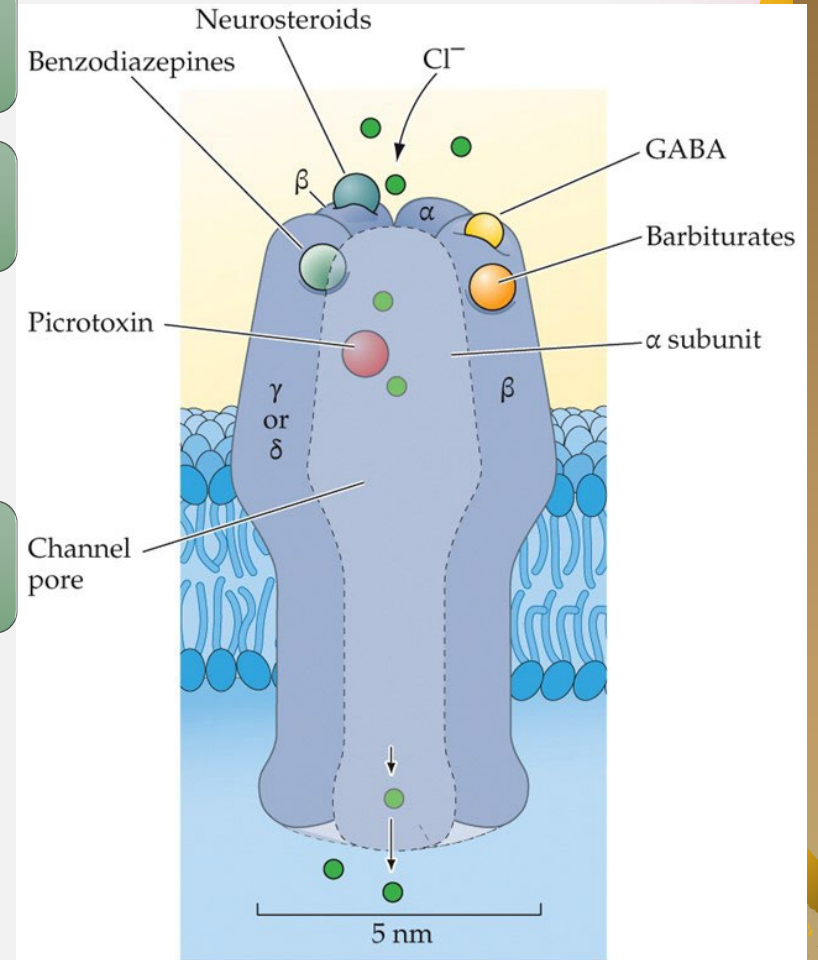
Ethanol

Volatile anesthetics

- propofol
- α_5 – amnestic

Neurosteroids

- allopregnanolone (Zulresso)
- Binding site on δ -subunit, converts GABA from partial agonist to full agonist



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GABA_B receptors are metabotropic

Postsynaptic GABA_B receptors

- Open K⁺ channels
- Inhibition of AC / ↓cAMP

Presynaptic GABA_B receptors

- Autoreceptors, heteroreceptors
- Inhibition of Ca²⁺ channels
- Inhibition of AC / ↓cAMP

